MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG3DSEMAT202-NUMERICAL METHODS

Duration: 2 hrs Maximum Marks: 70

*Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.*

**Part A**
**Answer any five questions. Each question carries 2 marks**

1. Find the relative error of the number 5.27 if both of its digits are correct. [E][1]
2. State the Newton-Raphson formula. Write any one advantage of Newton-Raphson method over bisection method. [K][2]
3. If a function is a polynomial of degree 4 , what will be its fifth forward difference? [An][4]
4. Write the general form of Gauss's backward interpolation formula. [K][4]
5. If $y_1 = 4, y_3 = 12, y_4 = 19$ and $y_x = 7$, find $x$. [A][3]
6. If a function is a polynomial of degree $n$, then what will be its $n$th divided difference? [U][4]
7. How does the rounding error behave with respect to step size $h$ in numerical differentiation? [U][1]
8. Is Simpson's 3/8 rule more accurate than Simpson's 1/3 rule? Why or why not? [An][4]

**Part B**
**Answer any five questions. Each question carries 6 marks**

9. Find a real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places using bisection method. [A][2]
10. Explain briefly the method of iteration to compute a real root of the equation $f(x) = 0$, stating the condition of convergence of the sequence of approximations. [U][2]
11. Show that $E \equiv 1 + \Delta$. [U][2]

12. Find the missing terms in the following table: [A][3]

| $x$ | 0 | 5 | 10 | 15 | 20 | 25 | 30 |
|---|---|---|---|---|---|---|---|
| $y$ | 1 | 3 | ? | 73 | 225 | ? | 1153 |

13. Find $y(2)$ from the following data using Lagrange's formula: [A][3]

| $x$ | 0 | 1 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $y$ | 0 | 1 | 81 | 256 | 625 |

14. Tabulate $y = x^3$ for $x = 2, 3, 4$ and 5, and calculate the cube root of 10 correct to three decimal places. [E][4]

15. Using Simpson's $\frac{1}{3}$ - rule with $h = 1$, evaluate the integral $I = \int_3^7 x^2 \log x \, dx$. [A][4]

16. Derive Simpson's $\frac{3}{8}$ - rule. [U][4]

**Part C**
**Answer any three questions. Each question carries 10 marks**

17. Find a real root of the equation $4e^{-x} \sin x - 1 = 0$ by regula - falsi method. Given that the root lies between 0 and 0.5. [A][2]

18. Using Newton's forward difference formula, find the sum $S_n = 1^3 + 2^3 + \cdots + n^3$. [A][4]

19. Using Gauss's forward formula, find the value of $f(32)$ given that $f(25) = 0.2707, f(30) = 0.3027, f(35) = 0.3386$ and $f(40) = 0.3794$. [A][4]

20. (a) Using the following table find $f(x)$ as a polynomial in $x$.

| $x$ | -1 | 0 | 3 | 6 | 7 |
|---|---|---|---|---|---|
| $f(x)$ | 3 | -6 | 39 | 822 | 1611 |

(b) Given $f(x) = \frac{1}{x^2}$. Find the divided differences $[a, b]$ and $[a, b, c]$. [A][3]

21. Assuming that the function values given in the following table are correct to the accuracy given, estimate the errors in the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.6$. [E][1]

| $x$ | 1.0 | 1.2 | 1.4 | 1.6 | 1.8 | 2.0 | 2.2 |
|---|---|---|---|---|---|---|---|
| $y$ | 2.7183 | 3.3201 | 4.0552 | 4.9530 | 6.0496 | 7.3891 | 9.0250 |

22. Evaluate (a) $\int_0^\pi x \sin x dx$ and (b) $\int_{-2}^2 \frac{x}{5+2x} dx$ using the trapezoidal rule with five ordinates. [E][4]